

1. Consider the table giving values for the forward and inverse functions.

$$f(6) = 5$$

$$f^{-1}(0) = -4$$

$$f(-5) = -4$$

$$f(0) = 1$$

$$f^{-1}(6) = 1$$

$$f^{-1}(-5) = 5$$

Find the values of the following.

$$f(5) = \boxed{}$$

$$f^{-1}(5) = \boxed{}$$

$$f(-4) = \boxed{}$$

$$f^{-1}(-4) = \boxed{}$$

$$f(1) = \boxed{}$$

$$f^{-1}(1) = \boxed{}$$

2. For many species of fish, the weight W is a function of the length x , given by $W = kx^3$, where k is a constant depending on the species. Suppose $k = 0.008$, W is in pounds, and x is in inches, so the weight is $W(x) = 0.008x^3$.

- Find the inverse function of this function.
- What does the inverse function give?
- Use the inverse function to find the length of a fish that weighs 64 pounds.
- In the context of the application, what are the domain and range of the inverse function?

- a. Give the inverse function.

$$W^{-1}(x) = \boxed{}$$

- b. What does the inverse function give? Choose the correct answer below.

- ☐ A. Given the length, the inverse function calculates the species constant, k .
- ☐ B. Given the length, the inverse function calculates the weight.
- ☐ C. Given the weight, the inverse function calculates the length.
- ☐ D. Given the weight, the inverse function calculates the species constant, k .

- c. A fish that weighs 64 pounds is inches long.

- d. Choose the correct domain and range of the inverse function below.

- ☐ A. The domain is $x < 0$ and the range is $W^{-1}(x) < 0$.
- ☐ B. The domain is $x < 0$ and the range is $W^{-1}(x) > 0$.
- ☐ C. The domain is $x > 0$ and the range is $W^{-1}(x) > 0$.
- ☐ D. The domain is $x > 0$ and the range is $W^{-1}(x) < 0$.